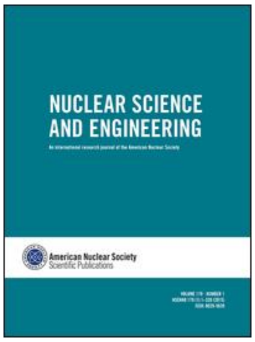

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Generalized Quasi-Static Method for Nuclear Reactor Space-Time Kinetics

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A generalization of the classical quasi-static method is described. "Amplitude" and "shape" functions are treated on the same footing by applying an iterative Newton method to the integral form of the multigroup diffusion equations. Provision is made for one amplitude function per neutron group. It is shown, further, how this model encompasses Ott and Meneley's improved quasi-static algorithm.

INTRODUCTION

Among all methods available for reactor space-time dynamics, the improved quasi-static (IQS) method devised by Ott and Meneley has gained widespread interest.[1] It has been largely successful in the description of power transients both in fast and thermal reactors as reported, for instance, in Refs. 2 and 3. Up to now, however, little attention has been paid to the mathematical grounds of the quasi-static method.[4] The purpose of this paper is to introduce the quasi-static formalism into a framework [generalized quasi-static (GQS)] that allows both generalizations like multigroup amplitude-shape factorizations or spectral matching[5] and the use of more rigorous algorithms like Newton-SOR procedures to solve the complete set of nonlinear equations.

We first briefly describe the main steps of the formalism (cf., also Ref. 6). Starting from the usual multigroup-diffusion model in differential form, the evolution equations are cast into the integral form, with the introduction of a time-averaged diffusion operator.[5] In the next stage, the quasi-static splitting

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[1]B. I. HAUSS and W. E. KASTENBERG, *Nucl. Sci. Eng.*, **69**, 326 (1979).

[2]K. O. OTT and D. A. MENELEY, *Nucl. Sci. Eng.*, **36**, 402 (1969).

[3]H. L. DODDS, *Nucl. Sci. Eng.*, **59**, 271 (1976).

[4]J. DEVOOGHT, *Ann. Nucl. Energy*, **7**, 47 (1980).

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[6]J. DEVOOGHT et al., *Proc. ANS/ENS Topl. Mtg. Nuclear Power Reactor Safety*, Brussels, Belgium, October 16-19, 1978, **3**, 2707 (1978).

 0029-5639/80/0010-0010$02.00/0

of each neutron group flux into the product of an amplitude function by a slowly varying shape function is performed, leading to a set of coupled nonlinear integral equations. These equations are treated on the same footing; the integral terms are eliminated using interpolation formulas, which yield usual one-step algorithms for the solution of initial value problems.[7] Provision is made for "spectral matching," which allows us to incorporate some known properties of the multigroup diffusion operator.[5,8] Finally, the resulting system of nonlinear equations is solved using the Newton-SOR technique.[9] This general approach can be shown to include Ott and Meneley's IQS algorithm as a particular case.

THE GENERALIZED QUASI-STATIC METHOD

A System of Coupled Integral Equations for Amplitude and Shape

We start from the time-dependent multigroup diffusion equations with G neutron groups and I delayed-neutron families:

$$\frac{1}{v}\frac{\partial \boldsymbol{\phi}(\boldsymbol{r},t)}{\partial t} = [-M(\boldsymbol{r},t) + F^p(\boldsymbol{r},t)]\boldsymbol{\phi}(\boldsymbol{r},t) + \sum_{i=1}^{I} \lambda_i \boldsymbol{\chi}_i^d C_i(\boldsymbol{r},t) \quad , \tag{1a}$$

$$\boldsymbol{\chi}_i^d \frac{\partial C_i(\boldsymbol{r},t)}{\partial t} = -\lambda_i \boldsymbol{\chi}_i^d C_i(\boldsymbol{r},t) + F_i^d(\boldsymbol{r},t)\boldsymbol{\phi}(\boldsymbol{r},t) \quad , \quad i = 1, \ldots, I \quad , \tag{1b}$$

where

$\boldsymbol{\phi}(\boldsymbol{r},t)$ and $C_i(\boldsymbol{r},t)$ = multigroup fluxes and delayed-neutron precursors

$M(\boldsymbol{r},t)$ = $G \times G$ lower triangular multigroup diffusion operator (we assume no upscattering, which is not essential)

$F^p(\boldsymbol{r},t)$ and $F_i^d(\boldsymbol{r},t)$ = prompt- and delayed-fission operators, respectively.

In this case

$$F^p(\boldsymbol{r},t) = (1-\beta)\boldsymbol{\chi}^p \cdot \boldsymbol{\nu\Sigma}_f(\boldsymbol{r},t)^T \quad , \quad \left(\beta = \sum_{i=1}^{I} \beta_i\right) \quad , \tag{2a}$$

$$F_i^d(\boldsymbol{r},t) = \beta_i \boldsymbol{\chi}_i^d \cdot \boldsymbol{\nu\Sigma}_f(\boldsymbol{r},t)^T \quad , \tag{2b}$$

where $\boldsymbol{a}^T$ is the row vector, transport of $\boldsymbol{a}$. Prompt-neutron spectra $\boldsymbol{\chi}^p$ generally differ from delayed-neutron spectra $\boldsymbol{\chi}_i^d$. Additionally, boundary conditions are specified at the outer edge of the reactor and an initial condition fixes the reactor power level at the start of the transient, i.e., $\boldsymbol{\phi}(\boldsymbol{r},0)$ and $C_i(\boldsymbol{r},0)$, $i = 1, \ldots, I$.

In shorthand notation, Eqs. (1a) and (1b) can be written

$$\frac{d\boldsymbol{y}(t)}{dt} = \Omega \boldsymbol{y}(t) + [A(t) - \Omega]\boldsymbol{y}(t) \tag{3}$$

with

$$\boldsymbol{y}(t) = [\boldsymbol{\phi} C_1 \ldots C_I]^T \quad ,$$

where Ω is some suitable "average" value of the multigroup-diffusion operator with delayed neutron precursors, $A(t)$. Integrating Eq. (3) formally over time scale t with $[A(t) - \Omega]\boldsymbol{y}(t)$ as a source, one obtains an expression that lends itself very easily to numerical computations:

$$\boldsymbol{y}(t) = \exp(t\Omega)\boldsymbol{y}(0) + \int_0^t ds \exp[(t-s)\Omega] \times [A(s) - \Omega]\boldsymbol{y}(s) \quad . \tag{4}$$

Any quadrature scheme for the right side of Eq. (4) yields an approximate value $\boldsymbol{y}(t)$.

Given the particular structure of Eqs. (1a) and (1b), the delayed-neutron precursors can be eliminated from Eqs. (1a) using (1b), so that the following relations are obtained from Eq. (4):

$$\begin{aligned}\boldsymbol{\phi}(\boldsymbol{r},t) &= \exp(t\Omega)\boldsymbol{\phi}(\boldsymbol{r},0) + \int_0^t ds \exp[(t-s)\Omega] \\ &\times \left\{ v[-M(s) + F^p(s)] - \Omega + \sum_i \lambda_i(\Omega+\lambda_i)^{-1} v F_i^d(s) \right\} \boldsymbol{\phi}(\boldsymbol{r},s) \\ &- \sum_i \lambda_i (\Omega+\lambda_i)^{-1} \int_0^t ds \exp[-\lambda_i(t-s)] \\ &\times v[F_i^d(s)\boldsymbol{\phi}(\boldsymbol{r},s) - \exp(\Omega s) F_i^d(0)\boldsymbol{\phi}(\boldsymbol{r},0)] \\ &+ \sum_i \lambda_i(\Omega+\lambda_i)^{-1} v[\exp(\Omega t) - \exp(-\lambda_i t)] \\ &\times [\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},0) - (\Omega+\lambda_i)^{-1} F_i^d(0)\boldsymbol{\phi}(\boldsymbol{r},0)] \quad ,\end{aligned} \tag{5a}$$

$$\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},t) = \exp(-\lambda_i t)\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},0) + \int_0^t ds \exp[-\lambda_i(t-s)] \times F_i^d(s)\boldsymbol{\phi}(\boldsymbol{r},s) \quad . \tag{5b}$$

These two integral equations are the starting point of the GQS formalism that follows.

We now introduce a splitting of each multigroup

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[7]C. W. GEAR, *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, Inc., Englewood Cliffs, New Jersey (1971).

[8]J. DEVOOGHT, *Nucl. Sci. Eng.*, **67**, 147 (1978).

[9]J. ORTEGA and W. RHEINBOLDT, *Iterative Solution of Nonlinear Equations in Several Variables*, Academic Press, Inc., London (1970).

flux $\phi_g(\boldsymbol{r},t)$ into the product of an amplitude function $T_g(t)$ by a mildly time-dependent shape function $\psi_g(\boldsymbol{r},t)$. Note that, in contrast to previous quasi-static approximations, each neutron group in this approach has *its own* amplitude function.[2,10] Some difference might then occur between the group amplitudes for transients in reactors having both thermal and fast fission, due to the spread in neutron generation times, as for instance in coupled fast-thermal reactors. For convenience, two different but equivalent notations will be used:

$$\boldsymbol{\phi}(\boldsymbol{r},t) = \begin{cases} \text{diag } \psi(\boldsymbol{r},t)\cdot \boldsymbol{T}(t) & \text{(6a)} \\ \text{diag } T(t)\cdot \boldsymbol{\psi}(\boldsymbol{r},t) \quad , & \text{(6b)} \end{cases}$$

where, for instance diag $\psi(\boldsymbol{r},t)$ is a $G \times G$ diagonal matrix with components $\psi_g(\boldsymbol{r},t)$, $g = 1, \ldots, G$, yielding, respectively, the point kinetics and the shape function equations.

To write the point kinetics equations, we still need a set of weight functions $W_g(\boldsymbol{r},t)$, $g = 1, \ldots, G$ and a normalization condition for $\boldsymbol{\psi}(\boldsymbol{r},t)$ to ensure uniqueness of the splitting of Eq. (6). Among the possible choices of weight functions, the adjoint *critical fluxes* still seem to remain the most adequate, although some attempts have been made to use time-dependent weight functions.[11] We, therefore, take the following vector of weight functions

$$\boldsymbol{W}(\boldsymbol{r}) = [\phi_1^*(\boldsymbol{r},0)\phi_2^*(\boldsymbol{r},0)\phi_G^*(\boldsymbol{r},0)]^T \quad , \tag{7}$$

where $\phi_g^*(\boldsymbol{r},0)$ is the normalized critical adjoint flux in group g.

Furthermore, we impose the condition that

$$\int_R d\boldsymbol{r}\phi_g^*(\boldsymbol{r},0)v_g^{-1}\psi_g(\boldsymbol{r},t) = 1 \quad , \quad \forall t > 0 \quad , \quad g = 1, \ldots, G, \tag{8}$$

with the integration being carried over the entire reactor volume. Introducing some straightforward generalizations of the classical point kinetics physical quantities, we define

$$1.\ F(t) = \langle \boldsymbol{W},(F^p + F^d)\psi\rangle_R \quad , \quad F^d = \sum_{i=1}^{I} F_i^d \quad ,$$

$$F = F^d + F^p \quad ,$$

$$F_{gg'}(t) = \int_R d\boldsymbol{r}\phi_g^*(\boldsymbol{r},0)[F^p(t) + F^d(t)]_{gg'}\psi_{g'}(\boldsymbol{r},t) \quad , \quad g,g' = 1, \ldots, G \quad . \tag{9a}$$

$$2.\ \text{diag } \Lambda F(t) = \langle \boldsymbol{W},v^{-1}\psi\rangle_R \quad ,$$

$$\Lambda F_g(t) = \int_R d\boldsymbol{r}\phi_g^*(\boldsymbol{r},0)v_g^{-1}\psi_g(\boldsymbol{r},t) \equiv \langle\phi_g^*(\boldsymbol{r},0),\ v_g^{-1}\psi_g(\boldsymbol{r},t)\rangle_R \quad . \tag{9b}$$

$$3.\ \widetilde{\beta}_i F(t) = \langle \boldsymbol{W}, F_i^d(t)\psi\rangle_R \quad ,$$

$$\widetilde{\beta}_i F_{gg'}(t) = \int_R d\boldsymbol{r}\phi_g^*(\boldsymbol{r},0)F_{igg'}^d(t)\,\psi_{g'}(\boldsymbol{r},t) \equiv \langle\phi_g^*(\boldsymbol{r},0), F_{igg'}^d\psi_{g'}(\boldsymbol{r},t)\rangle_R \quad . \tag{9c}$$

$$4.\ \left[\rho(t) + \frac{k_0 - 1}{k_0}I - \widetilde{\beta}(t)\right]F(t) = \langle \boldsymbol{W},[-M + F^p]\,\psi\rangle_R \quad ,$$

$$\{[\ldots]F(t)\}_{gg'} = \int_R d\boldsymbol{r}\phi_g^*(\boldsymbol{r},0) \times [-M(t) + F^p(t)]_{gg'}\psi_{g'}(\boldsymbol{r},t) \quad . \tag{9d}$$

$$5.\ \boldsymbol{\eta}_i(t) = \langle \boldsymbol{W},\mathbf{X}_i^d C_i(\boldsymbol{r},t)\rangle_R \quad ,$$

$$\eta_{ig}(t) = \int_R d\boldsymbol{r}\phi_g^*(\boldsymbol{r},0)\chi_{i,g}^d C_i(\boldsymbol{r},t) \quad . \tag{9e}$$

The bracket notations were used in definitions in Eqs. (9a) through (9e); they must be regarded only as a convenient way to express spatial integration over the reactor volume. The value k_0 appearing in Eq. (9d) is the value of the multiplication constant, which ensures criticality at $t = 0$ (cf., Ref. 12), and I is the identity operator.

With $\boldsymbol{R}(t) \equiv \Lambda(t)F(t)\boldsymbol{T}(t)$ and given Eqs. (9a) through (9e), one gets readily from Eqs. (5a) and (5b) the multigroup point kinetics equations whose integral form may be written (in compact notation):

$$\begin{aligned} \boldsymbol{R}(t) = {} & \exp(t\Omega)\boldsymbol{R}(0) + \int_0^t ds \exp[(t-s)\Omega] \\ & \times \left\{\left[\rho(s) + \frac{k_0 - 1}{k_0}I - \widetilde{\beta}(s)\right]\Lambda^{-1}(s) - \Omega\right\}\boldsymbol{R}(s) \\ & + \sum_i \lambda_i(\Omega + \lambda_i)^{-1}\int_0^t ds \exp[(t-s)\Omega] \\ & \times \{1 - \exp[-(t-s)(\Omega + \lambda_i)]\}\ \widetilde{\beta}_i(s)\Lambda^{-1}(s)\boldsymbol{R}(s) \\ & + \sum_i \lambda_i(\Omega + \lambda_i)^{-1}[\exp(\Omega t) - \exp(-\lambda_i t)]\boldsymbol{\eta}_i(0) \end{aligned} \tag{10a}$$

and

$$\boldsymbol{\eta}_i(t) = \exp(-\lambda_i t)\boldsymbol{\eta}_i(0) + \int_0^t ds \exp[-\lambda_i(t-s)] \times \widetilde{\beta}_i(s)\Lambda^{-1}(s)\boldsymbol{R}(s) \quad , \quad i = 1, \ldots, I \quad . \tag{10b}$$

The solution of Eqs. (10a) and (10b) depends, of course, on the knowledge of $\psi(\boldsymbol{r},t)$ through the value

[10]A. F. HENRY, *Nucl. Sci. Eng.*, **3**, 52 (1958).

[11]M. BECKER, *Nucl. Sci. Eng.*, **31**, 458 (1968).

[12]D. A. MENELEY, K. O. OTT, and E. S. WIENER, "Fast-Reactor Kinetics–The QXI Code," ANL-7769, Argonne National Laboratory (1971).

of the neutronic constants given in Eqs. (9a) through (9e): reactivity, weighted delayed-neutron fractions, and prompt-neutron generation time.

In connection with the shape functions $\boldsymbol{\psi}(\boldsymbol{r},t)$, Eqs. (5a) and (5b) and the splitting Eq. (6b) yields in straightforward fashion

$$
\begin{aligned}
T(t)\boldsymbol{\psi}(\boldsymbol{r},t) &= \exp(t\Omega)T(0)\boldsymbol{\psi}(\boldsymbol{r},0) + \int_0^t ds \exp[(t-s)\Omega] \\
&\times \left\{ v[-M(s) + F^p(s)] - \Omega + \sum_i \lambda_i(\Omega+\lambda_i)^{-1} v F_i^d(s) \right\} \\
&\times T(s)\boldsymbol{\psi}(\boldsymbol{r},s) - \sum_i \lambda_i(\Omega+\lambda_i)^{-1} \int_0^t ds \exp[-\lambda_i(t-s)] \\
&\times v[F_i^d(s)\boldsymbol{T}(s)\boldsymbol{\psi}(\boldsymbol{r},s) - \exp(\Omega s)F_i^d(0)T(0)\boldsymbol{\psi}(\boldsymbol{r},0)] \\
&+ \sum_i \lambda_i(\Omega+\lambda_i)^{-1} v[\exp(\Omega t) - \exp(-\lambda_i t)] \\
&\times [\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},0) - (\Omega+\lambda_i)^{-1} F_i^d(0)T(0)\boldsymbol{\psi}(\boldsymbol{r},0)] \qquad (11a)
\end{aligned}
$$

and

$$
\begin{aligned}
\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},t) &= \exp(-\lambda_i t)\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},0) \\
&+ \int_0^t ds \exp[-\lambda_i(t-s)] F_i^d(s) T(s)\boldsymbol{\psi}(\boldsymbol{r},s) \quad . \qquad (11b)
\end{aligned}
$$

Similarly, to determine the shape functions $\boldsymbol{\psi}(\boldsymbol{r},t)$, one needs $T(t)$, which is itself given by Eqs. (10a) and (10b). In fact, up to now, nothing has been done but to transform the multigroup-diffusion equations into a set of coupled nonlinear integral equations that could lend itself to an iterative technique, hopefully convergent toward the exact solution of the problem.[4] To evaluate this solution numerically, we need also to introduce approximate algorithms for the integral terms in Eqs. (10) and (11).

Approximate Quadrature and Spectral Matching

To take full advantage of the quasi-static assumption concerning the time variation of the shape function, we introduce two time-scales with step lengths h_m and h_M. Here, h_M is an integer multiple of h_m. The latter applies to the point kinetics equations whereas the former will be used to invert the full space-time shape problem. Hopefully, $h_M \gg h_m$ at least inasmuch as this inequality is compatible with the required precision, i.e., in fact with the nature of the transient.

Various numerical schemes can be chosen for the integral terms of Eqs. (10a) through (11a) provided they lead to implicit A-stable algorithms. It is known for instance that osculatory polynomial interpolation at both ends of the integration step leads to the family of Padé algorithms, whose stability properties have been thoroughly studied (cf., for instance, Refs. 13, 14, and 15). Another suitable choice would consist of a Gaussian quadrature scheme such as Gauss, Radau, or Lobatto's algorithms from Refs. 16 and 17.

Let

$$\int_0^h ds A(s)\boldsymbol{\phi}(s) \qquad (12)$$

denote any of the integral terms in Eqs. (10) and (11); we approximate the integrand in the following manner:

$$A(s)\boldsymbol{\phi}(s) \simeq A(0)\boldsymbol{\phi}(0)\cdot g_1(s) + A(h)\boldsymbol{\phi}(h)g_2(s) \ , \qquad (13)$$

with $g_1(s)$, $g_2(s)$ two interpolation basis functions satisfying cardinality properties, i.e., in compact form, using Kronecker's δ:

$$g_i[(j-1)h] = \delta_{ij} \quad . \qquad (14)$$

Linear basis functions would lead to the Crank-Nicolson scheme; we choose instead a linear combination of two exponential functions with time constants μ_1 and μ_2:

$$g_1(s) = \frac{\exp(\mu_1 s + \mu_2 h) - \exp(\mu_1 h + \mu_2 s)}{\exp(\mu_2 h) - \exp(\mu_1 h)} \qquad (15a)$$

and

$$g_2(s) = \frac{\exp(\mu_2 s) - \exp(\mu_1 s)}{\exp(\mu_2 h) - \exp(\mu_1 h)} \quad . \qquad (15b)$$

The reason for this choice is the fact that if indeed the true solution of the problem has an exponential behavior with time constant μ_1—the algebraically dominant solution of the inhour equation—the approximate scheme resulting from Eq. (13) *yields the exact solution*. This is the property known as "spectral matching" (cf., Refs. 5, 18, and 19). It is easily recognized that for μ_1, $\mu_2 \to 0$, $g_1(s)$ and $g_2(s)$ tend to the linear functions:

$$
\begin{aligned}
g_1(s) &= 1 - \frac{s}{h} \ ; \\
g_2(s) &= \frac{s}{h} \quad . \qquad (16)
\end{aligned}
$$

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[14] J. P. HENNART, *Math. Comp.*, **31**, 24 (1977).

[15] B. L. EHLE, *SIAM J. Math. Anal.*, **4**, 671 (1973).

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[17] J. DEVOOGHT et al., "Development of New Numerical Methods for Space-Dependent Nuclear Reactor Dynamics," ECC DG-III-E2, Université Libre de Bruxelles (1977).

[18] J. DEVOOGHT and E. MUND, Personal Communication.

[19] W. LINIGER and R. A. WILLOUGHBY, *SIAM J. Numer. Anal.*, **7**, 47 (1970).

We first develop the structure of the multigroup point kinetics equations. To simplify, let

$$C(s) = \rho(s) + \frac{k_0 - 1}{k_0} I - \widetilde{\beta}(s) + \sum_j \lambda_j(\Omega + \lambda_j)^{-1}\widetilde{\beta}_j(s) \quad . \tag{17}$$

Substituting Eq. (17) into Eq. (10a) and rearranging terms leads to

$$\begin{aligned} \boldsymbol{R}(t) = \exp(\Omega t)\boldsymbol{R}(0) + \int_0^t ds \exp[\Omega(t-s)] \\ \times \Big\{ C(s) - \Omega\Lambda(s) - \sum_i \lambda_i(\Omega + \lambda_i)^{-1} \\ \times \exp[-(\Omega + \lambda_i)(t-s)]\widetilde{\beta}_i(s) \Big\} \Lambda^{-1}(s)\boldsymbol{R}(s) \\ + \sum_i \lambda_i(\Omega + \lambda_i)^{-1}[\exp(\Omega t) - \exp(-\lambda_i t)]\boldsymbol{\eta}_i(0) \quad . \end{aligned} \tag{18}$$

We now apply the expansions of Eqs. (13) and (15) to the integrand of Eq. (18), except for the exponential term $\exp[-(\Omega + \lambda_i)(t - s)]$, which can be treated explicitly. The multigroup point kinetics equations on time-scale h_m become, after some algebra,

$$\begin{aligned} G(h_m) \equiv S_2^*(h_m)\boldsymbol{T}(h_m) - S_1^*(0)\boldsymbol{T}(0) \\ - \sum_i \lambda_i(\Omega + \lambda_i)^{-1} [\exp(\Omega h_m) \\ - \exp(-\lambda_i h_m)]\boldsymbol{\eta}_i(0) = 0 \quad , \end{aligned} \tag{19a}$$

with delayed-neutron precursors updating

$$\begin{aligned} G_i(h_m) \equiv \boldsymbol{\eta}_i(h_m) - \exp(-\lambda_i h_m)\boldsymbol{\eta}_i(0) \\ - \int_0^{h_m} ds \exp[-\lambda_i(h_m - s)]g_1(s)\cdot\widetilde{\beta}_i(0)F(0)\boldsymbol{T}(0) \\ - \int_0^{h_m} ds \exp[-\lambda_i(h_m - s)]g_2(s)\cdot\widetilde{\beta}_i(h_m) \\ \times F(h_m)\boldsymbol{T}(h_m) = 0 \quad . \end{aligned} \tag{19b}$$

The matrix operators $S_1^*(0)$ and $S_2^*(h_m)$ introduced in Eq. (19a) have the following structure:

$$\begin{aligned} S_2^*(h_m) = I - \int_0^{h_m} ds \exp[\Omega(h_m - s)]g_2(s) \\ \times [C(h_m) - \Omega\Lambda(h_m)]\cdot F(h_m) \\ + \sum_i \lambda_i(\lambda_i + \Omega)^{-1}\cdot\int_0^{h_m} ds \exp[-\lambda_i(h_m - s)] \\ \times g_2(s)\cdot\widetilde{\beta}_i(h_m)F(h_m) \quad , \end{aligned} \tag{20a}$$

and

$$\begin{aligned} S_1^*(0) = \exp(\Omega h_m) + \int_0^{h_m} ds \exp[\Omega(h_m - s)]g_1(s) \\ \times [C(0) - \Omega\Lambda(0)]\cdot F(0) - \sum_i \lambda_i(\lambda_i + \Omega)^{-1} \\ \times \int_0^{h_m} ds \exp[-\lambda_i(h_m - s)]g_1(s) \\ \times \widetilde{\beta}_i(0)F(0) \quad . \end{aligned} \tag{20b}$$

The scheme (19a) has the interesting property of yielding the exact solution of the multigroup point kinetics equations for two different eigensolutions of the *time-independent* inhour equation (cf., Ref. 20, p. 237):

$$\begin{aligned} \det|C - \Omega\Lambda| = \det\Big|\rho + \frac{k_0 - 1}{k_0} I - \widetilde{\beta} \\ + \sum_i \lambda_i(\lambda_i + \Omega)^{-1}\widetilde{\beta}_i - \Omega\Lambda\Big| = 0 \quad . \end{aligned} \tag{21}$$

Let μ_1 and μ_2 satisfy Eq. (21) with eigenvectors $\boldsymbol{\psi}_1$ and $\boldsymbol{\psi}_2$. We claim that, using Eq. (19a) with $\Omega = \mu_j$ $(j = 1,2)$ and the corresponding initial conditions:

$$\boldsymbol{T}(0) = \boldsymbol{\psi}_j \tag{22a}$$

and

$$\boldsymbol{\eta}_i(0) = \frac{\widetilde{\beta}_i}{\lambda_i + \mu_j} F(0)\boldsymbol{T}(0) \quad , \qquad j = 1,2 \quad , \tag{22b}$$

we have

$$\boldsymbol{T}(h_m) = \exp(\mu_j h_m)\boldsymbol{T}(0) \quad . \tag{23}$$

This results from straightforward substitution of Eq. (22b) into Eq. (19a) and the use of the following integral relation between the basis functions, Eq. (15):

$$\begin{aligned} \int_0^{h_m} ds \exp[-\lambda_i(h_m - s)]\,[g_1(s) + \exp(\mu_j h_m)g_2(s)] \\ = \frac{\exp(\mu_j h_m) - \exp(-\lambda_i h_m)}{\lambda_i + \mu_j} \quad . \end{aligned} \tag{24}$$

It has been shown numerically that "spectral matching" of the algebraically largest root of Eq. (21) and of the eigenvalue corresponding to prompt neutron generation for reactivity steps leads to quite accurate results and is particularly efficient for very stiff problems such as those encountered in the study of fast reactor transients (see Refs. 5 and 18). One could argue that step-like perturbations are almost trivial and unrealistic. The formalism developed above, however, remains unchanged and keeps its attractiveness if one replaces the constant operator Ω by the average value

$$\langle\Omega\rangle = \frac{1}{h}\int_0^h ds\omega(s)\cdot I \quad , \tag{25}$$

where, for instance, $\omega(s)$ is the instantaneous algebraically largest root of Eq. (21).

Similar considerations must be applied to the shape function equations [Eqs. (11a) and (11b)] on the large h_M time-scale. Exponential basis functions $h_1(s)$ and $h_2(s)$ of the type in Eq. (15) are used to expand

$$\{v[-M(s) + F^p(s)] - \Omega\}\boldsymbol{\psi}(\boldsymbol{r},s)$$

and

$$F_i^d(s)\boldsymbol{\psi}(\boldsymbol{r},s) \quad , \tag{26}$$

with different time constants, however. Since, by definition, $\boldsymbol{\psi}(\boldsymbol{r},t)$ is almost time independent, one can adopt $\mu_1 \simeq 0$ in this case. Introducing the expansions of Eq. (26) into Eqs. (11a) and (11b) yields two similar expressions to those satisfied by the amplitude functions $\boldsymbol{T}$ [Eqs. (19) and (20)]:

$$H \equiv S_2(h_M)\boldsymbol{\psi}(\boldsymbol{r},h_M) - S_1(0)\boldsymbol{\psi}(\boldsymbol{r},0) - v\sum_i \lambda_i(\Omega + \lambda_i)^{-1}[\exp(\Omega h_M) - \exp(-\lambda_i h_M)] \times \boldsymbol{\chi}_i^d C_i(\boldsymbol{r},0) = 0 \quad , \tag{27a}$$

$$H_i \equiv \boldsymbol{\chi}_i^d C_i(\boldsymbol{r},h_M) - \exp(-\lambda_i h_M)\boldsymbol{\chi}_i^d C_i(\boldsymbol{r},0) - \int_0^{h_M} ds\alpha_1(s)T(s)\cdot F_i^d(h_M)\boldsymbol{\psi}(\boldsymbol{r},h_M) - \int_0^{h_M} ds\alpha_2(s)T(s)\cdot F_i^d(0)\boldsymbol{\psi}(\boldsymbol{r},0) = 0 \quad . \tag{27b}$$

The differential matrix operators $S_1(0)$ and $S_2(h_M)$ are given by:

$$S_1(0) = \exp(\Omega h_M)T(0) + \{v[-M(0) + F^p(0)] - \Omega\} \times \int_0^{h_M} ds\alpha_1(s)T(s) + v\sum_i \lambda_i F_i^d(0) \times \int_0^{h_M} ds\beta_{1,i}(s)T(s) \tag{28a}$$

and

$$S_2(h_M) = T(h_M) - \{v[-M(h_M) + F^p(h_M)] - \Omega\} \times \int_0^{h_M} ds\alpha_2(s)T(s) - v\sum_i \lambda_i F_i^d(h_M) \times \int_0^{h_M} ds\beta_{2,i}(s)T(s) \quad , \tag{28b}$$

with the auxiliary quantities

$$\alpha_j(s) = h_j(s)\exp[\Omega(h_M - s)]$$

and

$$\beta_{j,i}(s) = \alpha_j(s)\,\frac{1 - \exp[-\Omega + \lambda_i)(h_M - s)]}{\Omega + \lambda_i} \quad , \quad j = 1,2 \quad . \tag{29}$$

To invert the shape function equations, one still needs a discrete approximation of the diffusion operators in $M(\boldsymbol{r},0)$ and $M(\boldsymbol{r},h_M)$. This is easily obtained by superposing a spatial grid on the reactor, compatible with both internal and external boundaries and then by considering a finite difference scheme or alternately, a family of finite elements.[20]

Operator $S_2(h_M)$ is almost singular. Difficulties appear in the iterative solution of Eq. (27a), when the normalization conditions of Eq. (8) are preserved. The best procedure is the introduction of a pseudo-eigenvalue.[21] However, in the case of the GQS approximation, we must introduce G pseudo-eigenvalues and the problem is solved in a more rigorous way by the Anselone-Rall algorithm, which is given in detail elsewhere.[22]

The Newton-SOR Algorithm

We now introduce in each macrostep of length h_M, a vector $\boldsymbol{x}$ whose components are made of the amplitude functions $\boldsymbol{T}(kh_m)$, delayed-neutron precursors $\boldsymbol{\eta}_i(kh_m)$ at each microstep kh_m ($k = 1, \ldots, n$), and the shape functions $\boldsymbol{\psi}(nh_m)$, $C_i(nh_m)$ at the lower bound of the next macrostep:

$$\boldsymbol{x} = [\boldsymbol{T}(h_m)\boldsymbol{\eta}_i(h_m)\boldsymbol{T}(2h_m)\boldsymbol{\eta}_i(2h_m) \ldots \boldsymbol{T}(nh_m)\boldsymbol{\eta}_i(nh_m)\boldsymbol{\psi}(nh_m)C_i(nh_m)]^T \quad . \tag{30}$$

The basic equations of Eq. (1) have been transformed into

$$\mathscr{F}(\boldsymbol{x}) = 0 \quad , \tag{31}$$

where $\mathscr{F}$ is a *nonlinear* algebraic operator defined by Eqs. (19), (20), (27), and (28), i.e.,

$$\mathscr{F}(\boldsymbol{x}) \equiv [G(kh_m)\ , \ G_1(kh_m) \ldots G_I(kh_m)\ , \ H\ , \ H_1 \ldots H_I]^T = 0 \quad . \tag{32}$$

To solve Eq. (31), we define a series of iterates $\boldsymbol{x}_m$, $m = 1, \ldots$ of the classical Newton-Raphson method and we expand $\mathscr{F}(\boldsymbol{x})$ linearly around $\boldsymbol{x}_m$:

$$\mathscr{F}(\boldsymbol{x}) \simeq \mathscr{F}(\boldsymbol{x}_m) + \mathscr{F}'(\boldsymbol{x}_m)(\boldsymbol{x} - \boldsymbol{x}_m) \quad , \tag{33}$$

where $\mathscr{F}'(\boldsymbol{x}_m)$ denotes the Fréchet derivative of $\mathscr{F}$ with respect to $\boldsymbol{x}_m$, i.e., of $G(kh_m)$, $G_i(kh_m)H$, and H_i with respect to the components of the m'th iterate of the unknown vector $\boldsymbol{x}$ (Ref. 23). The expressions of the components of $\mathscr{F}'(\boldsymbol{x}_m)$, which

[20]J. P. HENNART and E. MUND, *Nucl. Sci. Eng.*, **62**, 55 (1977).

[21]D. R. FERGUSON et al., "FX2, A Quasistatic Multidimensional Multigroup Diffusion Theory Code with Feedback," FRA-TM-87, Argonne National Laboratory (1976).

[22]J. DEVOOGHT, *ANS Topl. Mtg. Computational Methods in Nuclear Engineering*, Williamsburg, Virginia, April 23-25, 1979, **3**, 135, American Nuclear Society (1979).

[23]L. B. RALL, *Computational Solution of Nonlinear Operator Equations*, John Wiley and Sons, Inc., New York (1969).

are very lengthy, are given in detail in Ref. 17. The Newton-SOR method is really the classical SOR iteration technique used in conjunction with the local linearization of Newton's method (cf., Ref. 9, p. 215). One introduces therefore a second index k for the SOR iterations and setting $\boldsymbol{x}_m^{(0)} \equiv \boldsymbol{x}_m$, the algorithm runs

$$\boldsymbol{x} = \boldsymbol{x}_m - [\mathscr{F}'(\boldsymbol{x}_m)]^{-1} \cdot \mathscr{F}(\boldsymbol{x}_m) \quad ,$$

i.e., at the k'th iteration of Eq. (33) [cf., Ref. 9, Eq. (12), p. 216],

$$\boldsymbol{x}_m^{(k)} = \boldsymbol{x}_m^{(0)} - \omega_m [H_m^{k-1} + \ldots I] \times [D_m - \omega_m L_m]^{-1} \mathscr{F}(\boldsymbol{x}_m) \quad , \tag{34}$$

with the usual matrix decomposition

$$\mathscr{F}'(\boldsymbol{x}_m) = D_m - L_m - U_m \tag{35}$$

and

$$H_m = (D_m - \omega_m L_m)^{-1} [(1 - \omega_m) D_m + \omega_m U_m] \quad . \tag{36}$$

Here, ω_m is a relaxation parameter chosen to enhance convergence at the m'th Newton iteration. Both iterations should be applied indefinitely to obtain the exact solution. This is not the case however in practice where only one SOR iteration is performed per Newton iteration. In that case, $\boldsymbol{x}_m^{(1)} \equiv \boldsymbol{x}_{m+1}$, and Eq. (34) becomes

$$\boldsymbol{x}_{m+1} = \boldsymbol{x}_m - \omega_m (D_m - \omega_m L_m)^{-1} \cdot \mathscr{F}(\boldsymbol{x}_m) \quad . \tag{37}$$

A further simplification results from putting $\omega_m = 1$, drastically reducing the computer storage requirements by canceling at iteration $(m + 1)$ the contribution of terms arising through the Fréchet derivatives from iteration (m). Although theorems of the convergence of Eq. (37) are available,[9] their practical implementation is difficult and has not been done; however, no difficulty has been observed in practice.[6]

Comparison with the IQS Algorithm

The IQS algorithm of Ott and Meneley uses the same splitting Eq. (6), but with one amplitude function only, common to all neutron groups, and one normalization condition. The point kinetics equations are solved using Volodka's parabolic collocation method,[24] which for constant operators reduces to the Padé(2,2) algorithm. The shape functions, $\boldsymbol{\psi}(\boldsymbol{r},t)$, satisfy the partial differential equation

$$\frac{1}{v}\left[\frac{dT}{dt}\boldsymbol{\psi}(\boldsymbol{r},t) + T(t)\frac{\partial \boldsymbol{\psi}(\boldsymbol{r},t)}{\partial t}\right] = [-M(\boldsymbol{r},t) + F^p(\boldsymbol{r},t)]\, T(t)\boldsymbol{\psi}(\boldsymbol{r},t) + \sum_i \chi_i^d \lambda_i C_i(\boldsymbol{r},t) \quad . \tag{38}$$

[24]L. VOLODKA, "Program MOVER," ANL-7310, p. 492, Argonne National Laboratory (1968).

The time derivative of $\boldsymbol{\psi}(\boldsymbol{r},t)$ is approximated using a simple backward difference

$$\frac{\partial \boldsymbol{\psi}(\boldsymbol{r},t)}{\partial t} \simeq \frac{\boldsymbol{\psi}(\boldsymbol{r},t) - \boldsymbol{\psi}(\boldsymbol{r},t - \Delta t)}{\Delta t} \quad , \tag{39}$$

which, using time-scale $(0,h_M)$ reduces Eq. (38) to

$$\left\{\left[T(h_M) + h_M \frac{dT}{dt}(h_M)\right] + h_M v [M(h_M) - F^p(h_M)]\, T(h_M)\right\} \boldsymbol{\psi}(\boldsymbol{r},h_M) = T(h_M)\boldsymbol{\psi}(\boldsymbol{r},0) + h_M v \sum_i \chi_i^d \lambda_i C_i(\boldsymbol{r},h_M) \quad ,$$

or, alternately,

$$\left(T(h_M) - h_M\left\{v[M(h_M) - F^p(h_M)] + \frac{dT(h_M)}{dt}\right\}\right)\boldsymbol{\psi}(\boldsymbol{r},h_M) = T(h_M)\boldsymbol{\psi}(\boldsymbol{r},0) + h_M v \sum_i \chi_i^d \lambda_i C_i(\boldsymbol{r},h_M) \quad . \tag{40}$$

This equation, when compared with Eqs. (27a) through (28), shows a recognizably peculiar structure. Let us, for instance, identify $\omega(s)$ in Eq. (25) with $(d/dt)\ln T(t)$. One consequently has

$$\exp(\Omega h_M) = \frac{T(h_M)}{T(0)} \quad . \tag{41}$$

We also introduce a set of basis functions that provide spectral matching at both ends of the negative real axis (i.e., $\mu_1 = 0$, $\mu_2 = -\infty$). Putting these values into Eqs. (15) and (29) gives

$$h_1(s) = 0 \quad ,$$

$$h_2(s) = 1 \quad ,$$

$$\alpha_1(s) = 0 \quad ,$$

and

$$\beta_{1,j}(s) = 0 \quad , \tag{42a}$$

whereas

$$\alpha_2(s) = \frac{T(h_M)}{T(s)} \quad ;$$

$$\beta_{2,j}(s) = \int_s^{h_M} dt\, \frac{T(h_M)}{T(s)} \exp[-\lambda_i(t - s)] \quad . \tag{42b}$$

Combining Eq. (42) with Eq. (28) yields

$$S_1(0) = T(h_M) \quad , \tag{43a}$$

$$S_2(h_M) = T(h_M) - \left\{v[-M(h_M) + F^p(h_M)] - \frac{1}{h_M}\ln\frac{T(h_M)}{T(0)}\right\} T(h_M) h_M - v \sum \lambda_i F_i^d(h_M) \times \int_0^{h_M} T(s)ds \int_s^{h_M} \frac{T(h_M)}{T(t)} \times \exp[-\lambda_i(t - s)]\, dt \quad . \tag{43b}$$

Comparing Eq. (40) with the above equations [Eqs. (43a) and (43b) coupled with Eq. (27a)], one recognizes the same equation except for the addition of a source term of order h_M^2. Therefore, the IQS algorithm is a special case of the GQS algorithm developed above, provided $\Omega h_M \ll 1$.

CONCLUSIONS

The GQS formalism has been implemented into the CASSANDRE code developed jointly at Université Libre de Bruxelles and Centre d'Etude Nucleaire/SCK-Mol. Preliminary validation tests have been run on a linear ramp excursion in a small liquid-metal fast breeder reactor structure, the properties of which were taken from a draft of the Argonne National Laboratory FX2 report. The results of these tests have already been reported in Ref. 6; only a brief comment will be given here, saving more detailed analysis for a further publication.

Several calculations of the total power at various times up to a superprompt critical state were made, with different numbers of macrosteps and of Newton-SOR iterations in each macrostep. The influence of these iterations appeared to be important and the results showed conclusively that good convergence of the shape functions is essential for an accurate computation of the reactivity. Significant differences in the final values of the power, with respect to a reference solution, could be attributed to insufficient Newton-SOR iterations. Near prompt criticality, the amplitude of the fluxes is extremely sensitive to small inaccuracies of the reactivity, itself strongly dependent on a good convergence of the shape functions. Small differences were observed between the four group amplitudes in the GQS model, bracketing the unique value of the FX2 problem, although admittedly the test problem is not one where different amplitudes should have been expected.

To conclude, let us summarize the main differences of the generalized GQS formalism with the IQS approximation.

1. The factorization of amplitude and shape is done for every group, which leads to a multipoint kinetics problem with G unknowns instead of 1 unknown in IQS.
2. Shapes and amplitudes are treated on an equal footing [Eq. (6)].
3. The nonlinear system of $2(I + 1)$ equations for shapes and amplitudes is solved in a coherent fashion by a Newton-SOR method. This leads to an organization of the iterations different, say, from FX2. No substantial difference appears in the numerical work since the shape function equations [Eqs. (27a) and (38)] are substantially equivalent.
4. The quasi-singular equation [Eq. (27a)] is solved by an Anselone-Rall algorithm, taking into account the fact that G normalization conditions must be satisfied.
5. The time integration is made for both amplitude and shape functions by means of an interpolatory quadrature that allows for spectral matching instead of a collocation method.

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